

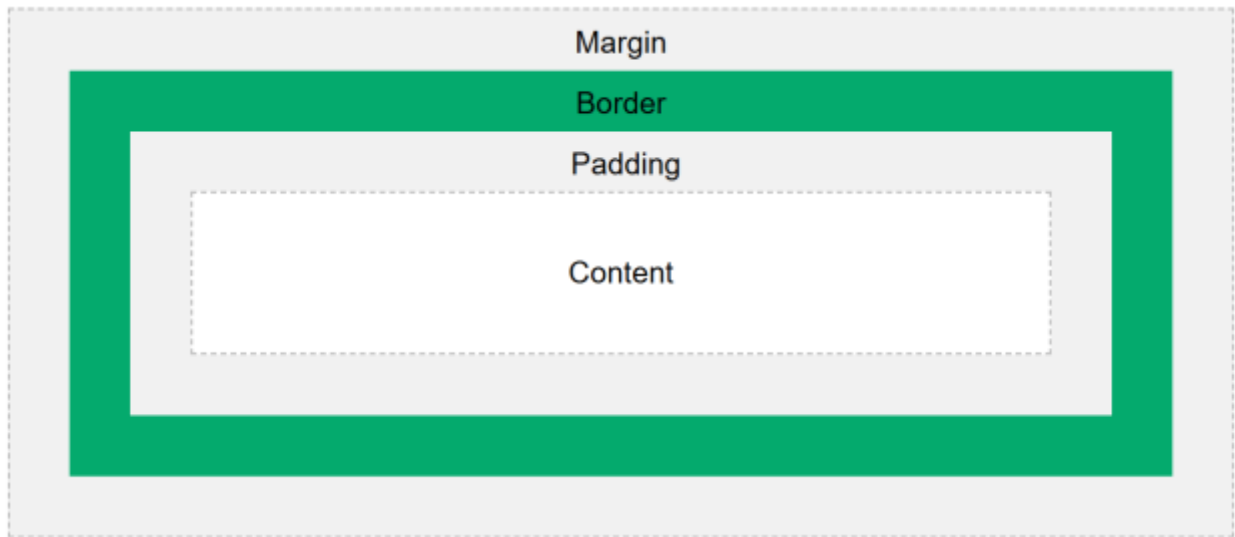
CSS Box Model (with Border Integration)

CSS Box Model defines how an element is **designed and spaced** using:

☞ **Content + Padding + Border + Margin**

Structure (Easy View)

Margin → Border → Padding → Content



1. Content

This is the **actual data** inside the element (text, image).

```
div {  
  width: 200px;  
  height: 100px;  
}
```

→ Only controls **inner content size**

2. Padding

Padding is the **space between content and border**.

```
div {  
  padding: 20px;  
}
```

☞ You can also write:

```
padding-top: 10px;  
padding-right: 20px;  
padding-bottom: 10px;  
padding-left: 20px;
```

→ Increases **inner spacing**

3. Border (Important – Your Topic Added Here)

Border surrounds padding and content

a) Shorthand Border

```
div {  
  border: 2px solid red;  
}
```

- 2px → width
 - solid → style
 - red → color
-

b) Border Width

```
div {  
  border-width: 3px;  
}
```

c) Border Style

```
div {  
  border-style: dashed;  
}
```

Common styles:

- solid
- dotted
- dashed
- double
- none

d) Border Color

```
div {  
    border-color: blue;  
}
```

e) Individual Borders

```
div {  
    border-top: 2px solid red;  
    border-right: 2px solid green;  
    border-bottom: 2px solid blue;  
    border-left: 2px solid black;  
}
```

f) Rounded Border

```
div {  
    border-radius: 10px;  
}
```

→ Makes **corners rounded**

4. Margin

Margin is the **space outside the border**.

```
div {  
    margin: 15px;  
}
```

☞ Individual:

```
margin-top: 10px;  
margin-right: 20px;  
margin-bottom: 10px;  
margin-left: 20px;
```

→ Controls **distance between elements**

CSS font-family Property

The `font-family` property is used to **set the typeface (font style) of text** in an HTML element.

It defines **which font should be used** to display the text.

Some Font Examples

Generic Font Family	Examples of Font Names
Serif	Times New Roman Georgia Garamond
Sans-serif	Arial Verdana Helvetica
Monospace	Courier New Lucida console Monaco
Cursive	<i>Brush Script MT</i> <i>Lucida Handwriting</i>
Fantasy	Copperplate Papyrus

Syntax

```
selector {  
  font-family: font-name;  
}
```

Example 1: Single Font

```
p {  
  font-family: Arial;  
}
```

→ The paragraph text will be displayed in **Arial font**.

CSS Icons – Font Awesome

Font Awesome is a popular **icon library** used in web development to add **icons (like home, user, search, etc.)** using **CSS and HTML**.

Instead of images, icons are added using **classes**, so they are **scalable and customizable**.

You can also get CDN from:

🔗 [cdnjs](#)

Search:

🔗 **"font awesome cdn cdnjs"**

Then copy the latest version link.

Where to Paste?

Paste inside `<head>` section:

```
<head>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">
</head>
```

Simple Understanding

Step	What to Do
1	Open Font Awesome site
2	Get CDN link
3	Paste in <code><head></code>
4	Use icons

1. How to Add Font Awesome

You must first **link Font Awesome** in your HTML.

Using CDN (Easy Method)

```
<head>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">
</head>
```

2. Using Icons

Icons are added using the `<i>` tag with specific classes.

```
<i class="fa fa-home"></i>
```

→ Displays a **home icon**

3. Common Icons Examples

```
<i class="fa fa-user"></i>      <!-- User icon -->
<i class="fa fa-search"></i>    <!-- Search icon -->
<i class="fa fa-envelope"></i>  <!-- Mail icon -->
<i class="fa fa-car"></i>      <!-- Car icon -->
```

4. Styling Icons using CSS

You can change **size, color, and style** using CSS.

Example:

```
i {
  color: red;
  font-size: 30px;
} → Icon becomes red and bigger
```

5. Different Sizes

Font Awesome provides size classes:

```
<i class="fa fa-home fa-2x"></i>
<i class="fa fa-home fa-3x"></i>
<i class="fa fa-home fa-5x"></i>
```

CSS Links (Hover Button & Link States)

CSS allows you to **style links** (<a> tag) in different states like hover, click, and visited.

◆ Four Link States

These are the **4 important pseudo-classes**:

```
a:link      { color: blue; }      /* Normal link */
a:visited   { color: purple; }    /* Visited link */
a:hover     { color: red; }       /* Mouse over */
a:active    { color: green; }     /* While clicking */
```

◆ Meaning (Easy Understanding)

State	Meaning
:link	Normal link (not visited)
:visited	Already clicked link
:hover	Mouse pointer over link
:active	When clicking the link

◆ Important Rule (Exam Point ⚠️)

☞ Order must be:

:link → :visited → :hover → :active

☞ Otherwise styles may not work correctly.

◆ Example Program

```
<!DOCTYPE html>
<html>
<head>
<style>
a:link {
    color: blue;
}

a:visited {
    color: purple;
}

a:hover {
    color: red;
}

a:active {
    color: green;
}
</style>
</head>

<body>

<a href="#">Click Me</a>

</body>
</html>
```


CSS Hover Button Color Change

To change the **button color when hovering**, we use the **:hover pseudo-class**.

◆ Basic Syntax

```
button:hover {  
    background-color: color;  
}
```

◆ Simple Example

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
button {  
    background-color: blue;  
    color: white;  
    padding: 10px 20px;  
    border: none;  
}  
  
button:hover {  
    background-color: red;  
}  
</style>  
</head>  
  
<body>  
  
<button>Click Me</button>  
  
</body>  
</html>
```

◆ Explanation

- Normal → **Blue button**
- On hover → **Turns Red**



1. `box-shadow` Explanation

Example 1:

```
box-shadow: 0 0 10px cyan, 0 0 20px cyan, 0 0 40px cyan;
```

☞ This creates a **strong glow effect**

Breakdown:

Each part:

```
0 0 10px cyan
```

Value	Meaning
0	Horizontal shadow (left/right)
0	Vertical shadow (up/down)
10px	Blur (spread of glow)
cyan	Color

Why 3 values?

```
0 0 10px cyan,  
0 0 20px cyan,  
0 0 40px cyan
```

☞ Multiple shadows = **more intense glow**

- 10px → small glow
- 20px → medium glow
- 40px → large glow

→ Together → **neon effect** 🔥

Example 2:

```
box-shadow: 0 0 10px red, 0 0 20px red;
```

☞ Less values → **weaker glow**

- Only 2 layers
- So glow is softer than above

2. `transition: 0.3s;`
`transition: 0.3s;`

☞ Makes changes **smooth**

1. Glow Button (Box Shadow)

```
<!DOCTYPE html>
<html>
<head>
<style>
button {
  background-color: black;
  color: white;
  padding: 10px 20px;
  border: none;
  transition: 0.3s;
}

button:hover {
  box-shadow: 0 0 10px red, 0 0 20px red;
}
</style>
</head>

<body>

<button>Glow Button</button>

</body>
</html>
```

→ When you hover → **red glow appears around button**

